

Two-panel material estimate

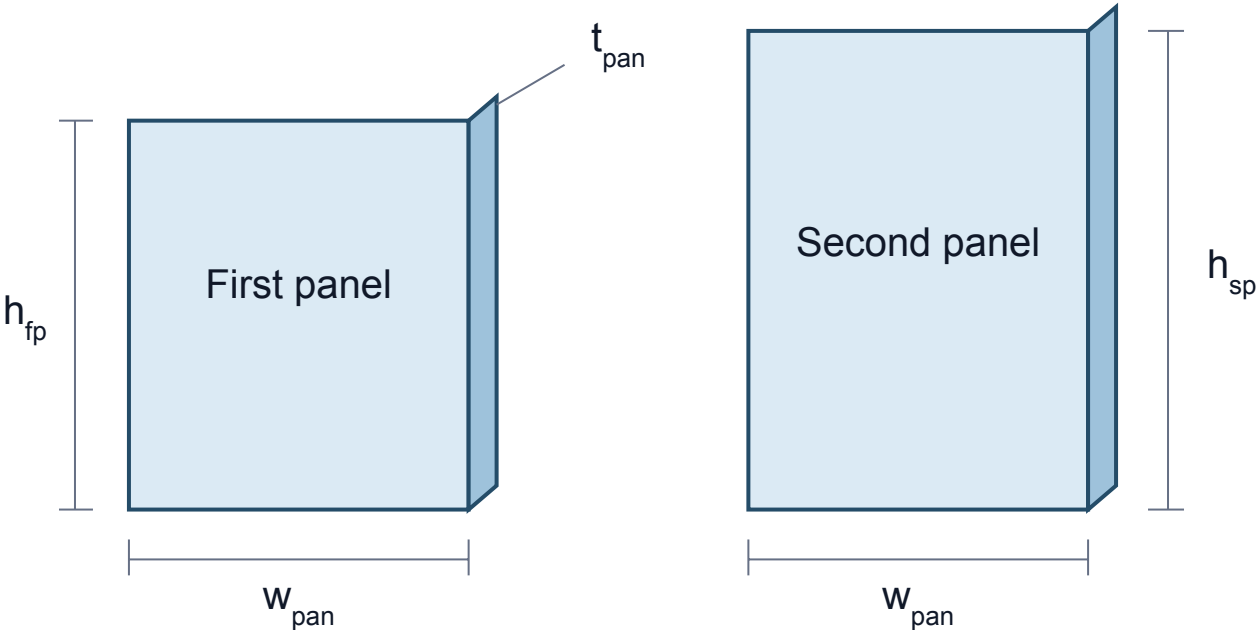
Execution-bound document. Numerical verification is reported separately.
Function estimate. Inputs width = 1, first_panel_height = 3, second_panel_height = 4, thickness = 0.1, density = 500.

Inputs and totals

0. Shared panel width	w_{pan}	1.00	m
1. First panel height	h_{fp}	3.00	m
2. Second panel height	h_{sp}	4.00	m
3. Uniform thickness	t_{pan}	0.10000	m
4. Uniform density	ρ_{mat}	500.00	$\frac{\text{kg}}{m^3}$

Rectangular panels; uniform thickness and density; no openings and no waste allowance. This is a material estimate, not a structural capacity calculation.

Symbol subscripts: pan = panel; fp = first panel; sp = second panel; rect = rectangle; tot = total; mat = material.



Two rectangular panels with shared width and uniform thickness.

Rectangle geometry

5. Panel area	$A_{\text{rect,fp}}$	3.00	m^2
$A_{\text{rect,fp}} = w_{\text{pan}} \cdot h_{\text{fp}} = 1.00 \cdot 3.00$			
6. Panel perimeter	$P_{\text{rect,fp}}$	8.00	m

$$P_{\text{rect,fp}} = 2 \cdot (w_{\text{pan}} + h_{\text{fp}}) = 2 \cdot (1.00 + 3.00)$$

Rectangle geometry

7. Panel area	$A_{\text{rect,sp}}$	4.00	m^2
$A_{\text{rect,sp}} = w_{\text{pan}} \cdot h_{\text{sp}} = 1.00 \cdot 4.00$			
8. Panel perimeter	$P_{\text{rect,sp}}$	10.00	m
$P_{\text{rect,sp}} = 2 \cdot (w_{\text{pan}} + h_{\text{sp}}) = 2 \cdot (1.00 + 4.00)$			
9. Total panel area	A_{tot}	7.00	m^2
$A_{\text{tot}} = A_{\text{rect,fp}} + A_{\text{rect,sp}} = 3.00 + 4.00$			

Material volume and mass

10. Material volume	V_{mat}	0.70000	m^3
$V_{\text{mat}} = A_{\text{tot}} \cdot t_{\text{pan}} = 7.00 \cdot 0.10000$			
11. Material mass	m_{mat}	350.00	kg
$m_{\text{mat}} = V_{\text{mat}} \cdot \rho_{\text{mat}} = 0.70000 \cdot 500.00$			